Linear Algebra II HW 6

Due Date: on or before 11:55 a.m., Mar/26/2025 (Wednesday), in T3-502-R26-H7.

1. In $C[0, 1]$, with inner product defined by Theorem 5.4.3, consider the vectors 1 and $x$.

(a) Find the angle $\theta$ between 1 and $x$.

(b) Determine the vector projection $\mathbf{p}$ of 1 onto $x$ and verify that $1 - \mathbf{p}$ is orthogonal to $\mathbf{p}$.

(c) Compute $\|1-\mathbf{p}\|, \|\mathbf{p}\|, \|1\|$ and verify that the Pythagorean law holds.

2. Let $A$ be a nonsingular $n \times n$ matrix and for each vector $\boldsymbol{x}$ in $\boldsymbol{R}^n$ define

$$\|\boldsymbol{x}\|_A = \|A\boldsymbol{x}\|$$

where $\|\boldsymbol{v}\|$ is defined in Definition 5.1.3. Show that the above equation defines a norm on $\boldsymbol{R}^n$.

3. Let $I_n$ be an $n\times n$ identity matrix and $0_n$ be an $n\times n$ zero matrix. Suppose $A$ is an $n\times n$ matrix such that $A^2 - 4A + 3I_n = 0_n$. Show that

$$N(A-3I_n) \oplus N(A-I_n) = \boldsymbol{R}^n.$$

4. Let $C^1[-1,1] = \{\text{all differentiable functions on } [-1,1] \text{ such that } f' \text{ is continuous on } [-1,1]\}$. For any $f, g \in C^1[-1,1]$, define

$$\langle f, g \rangle_1 = f(0)g(0) + \int_{-1}^{1} f'(x)g'(x)dx$$

(a) Show that $\langle\ ,\ \rangle_1$ is an inner product on $C^1[-1,1]$.

(b) Let $W_1 = \{f \in C^1[-1,1] : f(-x) = -f(x)\}$ and $W_2 = \{g \in C^1[-1,1] : g(-x) = g(x)\}$. Show that $W_1 \perp W_2$ under $\langle\ ,\ \rangle_1$.

(c) Show that $C^1[-1,1] = W_1 \oplus W_2$. Hence show that $W_1 = W_2^{\perp}$.

5. Let $x_1, \cdots, x_n$ be distinct real numbers. Show that the below equation defines an inner product on $P_n$ (Example 4.1.10) by

$$\langle p, q \rangle = \sum_{i=1}^{n} p(x_i) q(x_i).$$

6. Let $C^0([0, \pi], \square)$ be a vector space of real valued continuous function on $[0, \pi]$. Show that

$$\text{span}\{\sin x, \cos x\} \oplus \{f \in C^0([0, \pi], \square) : f(0) = f(\tfrac{\pi}{2}) = f(\pi)\} = C^0([0, \pi], \square).$$